

METHODS: MODELED EXTENT AND BOUNDARY STATISTICS

Modeled extent

The modeled extent is a rectangular bounding box enclosing the USGS GAP range polygon for the species after that polygon has been buffered outward by 250 km, computed in a projected equal-area coordinate system (EPSG:5070, USA Contiguous Albers Equal Area Conic). The buffered polygon is species-specific. The distance is not. The same 250 km is used for every species. This keeps the boundary condition standardized and avoids setting a parameter of the analysis from centroid velocity, which the analysis itself produces.

Derivation of the 250 km buffer

The figure derives from Huang, Sauer, and Dubayah (2017), who tracked the abundance-weighted geographic centroid of 57 permanent resident North American bird species over 44 years of Breeding Bird Survey data and found an average multidirectional centroid velocity of 5.89 km/yr. Projected across rENM's 45-year study window this gives approximately 265 km. Huang et al. report a 259 km average displacement directly, over a comparable period. The round figure of 250 km is slightly conservative against both.

Other bird range-shift estimates report only a poleward or latitudinal component of movement. The multidirectional figure was chosen because rENM computes a full two-dimensional suitability centroid, and the pilot species examined during development did not shift uniformly poleward.

What the buffer does and does not affect

Range-based summary statistics are computed against the unbuffered GAP range polygon, not the bounding box. This covers positive and negative trend percentage, hot spot area, and hot spot percentage. The buffer and its enclosing rectangle keep the trend surface from being clipped at the historic range edge. They do not redefine what counts as range in the reported tables.

The buffer ring is the zone between the raw GAP polygon and the 250 km buffered polygon. Trend behavior there is reported separately as a boundary trend statistic. Comparing the ring against the range interior shows whether conditions just outside the historic range are improving or deteriorating relative to the occupied range. A static range polygon cannot show this.

Limitations

The 250 km figure is drawn from permanent resident species. Huang et al. chose residents because a resident species' abundance centroid is a cleaner climate signal than a migrant's, whose breeding and wintering ranges can shift independently. The figure may not suit strongly migratory species.

The buffer distance is a provisional default. It should be revisited as more species run through the pipeline and as taxon- or guild-specific velocity estimates become practical to compute internally.

Reference

Huang, Q., J. R. Sauer, and R. O. Dubayah. 2017. Multidirectional abundance shifts among North American birds and the relative influence of multifaceted climate factors. *Global Change Biology* 23:3610–3622. <https://doi.org/10.1111/gcb.13683>